



National Comprehensive
Cancer Network®

2026

NCCN Guidelines for Patients®

Cancer care recommendations from leading experts at the
National Comprehensive Cancer Network® (NCCN®)

Rectal Cancer



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Available online at
[NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines)



NCCN Guidelines for Patients®

The essential guide for people facing cancer.

Based on care recommendations from leading cancer experts.

Explains high-quality cancer care provided at
state-of-the-art cancer centers.

Reviewed and revised every year.

Did you know that top cancer centers across the United States work together to improve cancer care? This alliance of leading cancer centers is called the National Comprehensive Cancer Network® (NCCN®).

Because cancer care is always evolving, NCCN develops and frequently updates evidence-based cancer care recommendations used by health care providers worldwide. These recommendations are known as the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®).

The NCCN Guidelines for Patients plainly explain these expert recommendations, so you can talk with your care team about the best care for you.

**These NCCN Guidelines for Patients are based on the
NCCN Guidelines® for Rectal Cancer, Version 2.2026 — April 7, 2026.**

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About rectal cancer

- 5 What is rectal cancer?
- 5 What are polyps?
- 6 How is rectal cancer treated?
- 7 How can I get the best care?

1 About rectal cancer

Rectal cancer is common and treatable. This guide describes treatment for adenocarcinoma, the most common type of rectal cancer.

The term colorectal cancer is used to describe cancers that form in either the colon or rectum. While these cancers are similar, their treatment is different. The focus of this guide is rectal cancer.

What is rectal cancer?

Rectal cancer is a disease that forms at the end of the large intestine. Cancer starts in the innermost layer that comes in contact with stool. If left untreated, cancer cells can grow into the deeper layers of the rectal wall and then into nearby areas of the body.

Cancer cells can also break off from the rectal tumor and travel through lymph or blood to nearby lymph nodes or to distant organs, like the liver or the lungs.

What are polyps?

Polyps are non-cancerous growths that form on the innermost lining of the intestine.

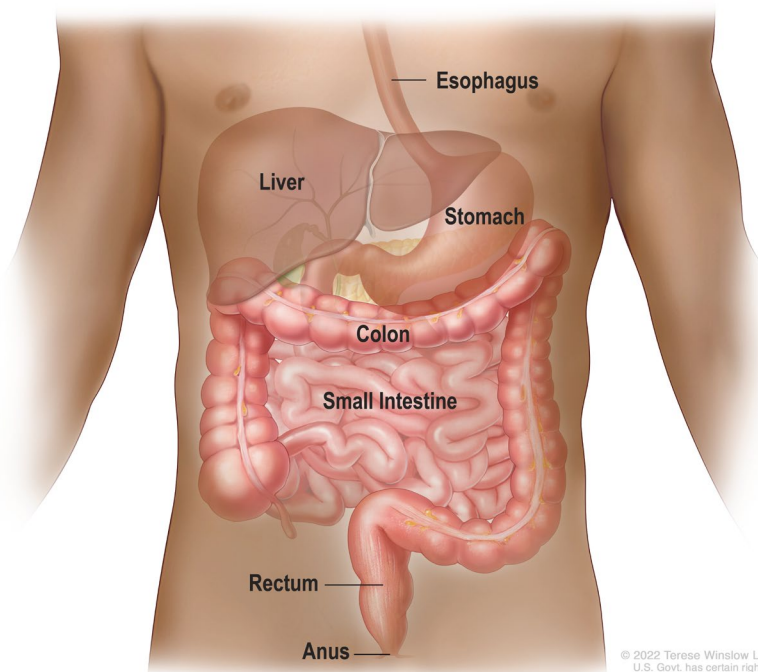
The most common type of polyp is called an adenoma. While it may take many years, adenomas can become invasive rectal cancer, called an adenocarcinoma. Adenocarcinoma is the most common type of rectal cancer.

Polyps that rarely turn into cancer include hyperplastic and inflammatory polyps.

Removing polyps can prevent cancer before it starts. Most can be removed during a colonoscopy using a minor endoscopic

The rectum

The rectum is the last 5 to 6 inches of the large intestine. After digested food becomes stool in the colon, it stays in the rectum until it exits the body through the anus.



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1 About rectal cancer

procedure called a polypectomy. Polyps can also be tested to make sure cancer hasn't already started to form.

For more information on finding polyps early, see *NCCN Guidelines for Patients: Colorectal Cancer Screening* at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



How is rectal cancer treated?

Surgery is recommended for early rectal cancers when possible. If the cancer hasn't reached the muscle layer of the rectal wall, less invasive options may include endoscopic submucosal dissection (ESD) and transanal local excision. Abdominal surgery is often recommended for cancers that have grown into or through the muscle layer, and for smaller cancers with high-risk features.

Locally advanced cancers have grown through the rectal wall or have spread to nearby lymph nodes. Treatment often involves external beam radiation therapy (EBRT) and chemotherapy. Or, for cancers with a feature called mismatch repair deficiency (dMMR), immunotherapy is recommended. Surgery will follow if needed and possible.

Stage 4 (metastatic) cancer is often treated with systemic therapy. Treatment with surgery or other local therapies may also be an option. The results of biomarker testing help your

What's in this resource?



Testing the tumor for an important feature called mismatch repair deficiency (dMMR)



Developing a care plan based on the cancer's mismatch repair (MMR) status



Treatment for polyps with cancer and all stages of rectal cancer



Survivorship

doctor decide which systemic therapies may work best. For dMMR metastatic cancers, immunotherapy is recommended if you are eligible.

How can I get the best care?

Advocate for yourself. You have an important role to play in your care. Many people feel more satisfied when they actively take part in planning their cancer care.

The NCCN Guidelines for Patients will help you play a larger role in your care. Discuss the recommendations in this guide with your care team. Ask questions about your options and share your goals and concerns.

Don't know what to ask? You're not alone. That's why we include suggested questions to ask at the end of chapters. Keep reading to find the best care for you.

How this guide can help you

Making decisions about cancer care is stressful. There's a lot to learn, and you don't know what the future holds.

Use this guide to get the information and support you need.

Patients, doctors, and other health care professionals trust the NCCN Guidelines for Patients. This guide uses clear, everyday language to explain current cancer care recommendations made by respected experts in the field. Their recommendations are based on the latest research and practices at leading cancer centers.

Your health is unique to you, so your cancer care should be, too. As you read this guide, you'll learn which treatments are likely to provide the best results for you. And you'll be better prepared to talk with your care team

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Testing for rectal cancer

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2 Testing for rectal cancer

Your care team will make a treatment plan just for you. First, they will do tests to learn about the cancer and your general health. All rectal cancers should be tested for mutations (changes) in genes that fix damaged DNA, called mismatch repair (MMR) genes.

If rectal cancer is suspected, testing as described in this chapter is recommended. Also see **Guide 1**.

Colonoscopy

Most colorectal cancers are found during colonoscopy. This procedure allows a health care provider, usually a gastroenterologist, to examine the entire large bowel.

If they see any polyps or abnormal areas of tissue, they will remove them during the procedure using a cutting tool. This is called a polypectomy. An expert called a pathologist will look for cancer cells in the polyp(s) using a microscope. If any are found, they will make a cancer diagnosis.

The diagnosis and other findings are included in a document called a pathology report. Ask

Guide 1

Tests to diagnose rectal cancer and make your treatment plan

Necessary tests

- Colonoscopy
- Biopsy
- Testing the tumor to determine mismatch repair (MMR) status
- Blood tests: Complete blood count (CBC), comprehensive metabolic panel (CMP), carcinoembryonic antigen (CEA)
- MRI of the pelvis to see the extent of cancer in and near the rectum
- CT of the chest and abdomen to check for cancer spread beyond the pelvis
- Testing the tumor for alterations in the PI3K pathway (to make decisions about taking aspirin after treatment to prevent recurrence)

Other tests or care you may receive

- Proctoscopy
- Endorectal ultrasound (if you can't have MRI)
- Testing for *DPYD* gene variants that cause DPD deficiency (if your treatment plan includes chemotherapy)
- Help to quit smoking
- Family planning and options for preserving fertility

2 Testing for rectal cancer

your provider for a copy or for instructions on how to access the results electronically.

For more information on screening colonoscopy and what to expect, see *NCCN Guidelines for Patients: Colorectal Cancer Screening* at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on [NCCN Patient Guides for Cancer](#) app.



Mismatch repair testing

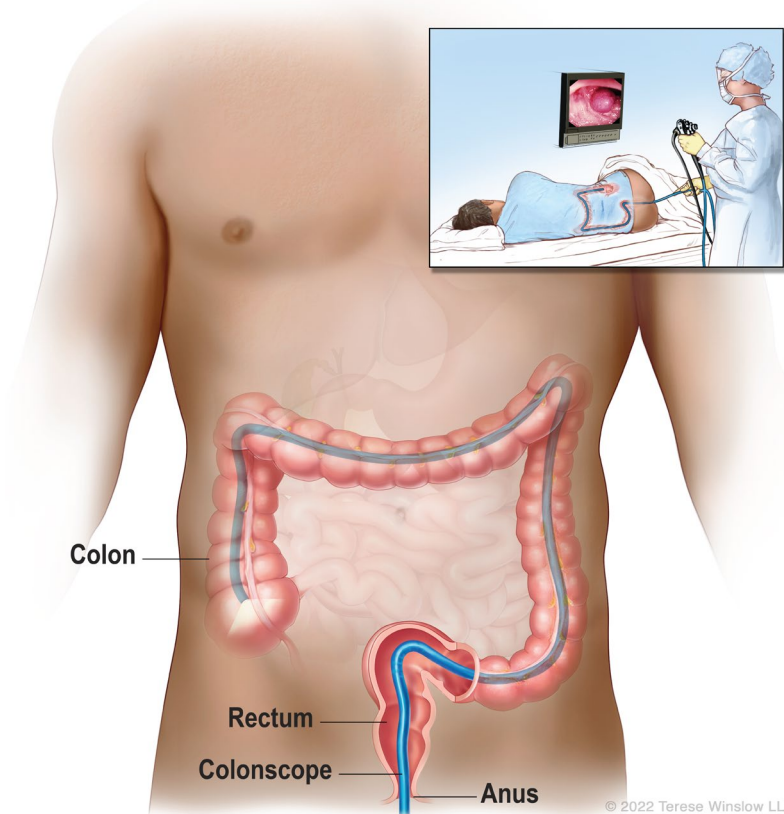
All rectal cancers should be tested for mutations (changes) in genes that normally find and fix damaged DNA, called mismatch repair (MMR) genes. Testing involves analyzing a piece of the rectal tumor in a lab.

Depending on the method used, an abnormal result is called either mismatch repair deficiency (dMMR) or microsatellite instability-high (MSI-H). These cancers have a high number of mutated (changed) genes and respond differently to rectal cancer treatments.

Most colorectal cancers don't have either of these features and are described as mismatch repair proficient (pMMR) or microsatellite stable (MSS).

Colonoscopy

Colonoscopy is a procedure that allows your doctor to see and remove any abnormal tissue from the large intestine. A thin device is inserted through the anus, up the rectum, and into the colon. The device has a light, a camera, and a cutting tool.



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Family health history

Your provider will ask about the health history of your biological (related by blood) family members. People with a first-degree relative with colorectal cancer are more likely to have this cancer than those without a family history of colorectal cancer.

Most colorectal cancers occur for unknown reasons. A small number of people are born with a disorder that makes them more likely to get colorectal and other cancers. These are called inherited cancer syndromes. Two such conditions include Lynch syndrome and familial adenomatous polyposis (FAP).

People born with Lynch syndrome are at high risk of developing colorectal, endometrial, ovarian, and several less common cancers. Lynch syndrome is caused by inherited mutations of the MMR genes. For cancers found to have the dMMR or MSI-H biomarker, testing for Lynch syndrome is recommended.

FAP causes many polyps to form in the colon and rectum. The polyps start as benign growths but can become invasive colorectal cancer over time. Cancer often develops by age 50 in people with FAP.

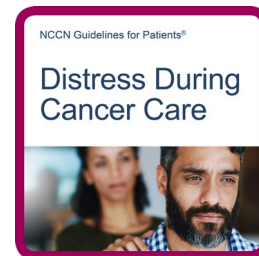
Genetic testing for hereditary cancer risk

Gene differences you are born with are called **germline variants**. If an inherited cancer or cancer syndrome is suspected, your provider may refer you to a genetic counselor. This expert can speak with you and your family about testing for syndromes related to colorectal cancer.

Distress

Depression, anxiety, fear, and distress are common after a cancer diagnosis. These feelings can make it harder to deal with cancer and its treatment. If you're worried, overwhelmed, or hopeless, ask your care team for help.

More information about cancer and distress is available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



To be tested, you must provide a sample of blood or saliva. A pathologist tests the sample for germline gene variants that cause these syndromes. Speaking with a genetic counselor is recommended for any positive genetic testing result.

PI3K pathway testing

Some rectal tumors have alterations in genes in the PI3K pathway. Genes in this pathway include *PI3KCA* and *PTEN*. Changes in these genes can activate an enzyme (PI3K) that causes the cancer to grow quickly and return after treatment.

Cancer experts have found that aspirin helps prevent recurrence (after recommended treatment) in cancers with these changes.

Confirm with your care team that the rectal tumor will be tested for somatic (non-hereditary) alterations that activate PI3K.

DPYD testing

Capecitabine and fluorouracil (5-FU) are chemotherapy medicines called fluoropyrimidines. They are often used to treat colorectal cancer. Your body uses a special enzyme (protein) called DPD to break down these medicines so they can work properly and leave your body safely and efficiently.

Around 1 in 15 people inherit a difference in the gene that makes DPD, the *DPYD* gene. This leads to less DPD protein being made, called DPD deficiency. In these people, fluoropyrimidines can build up to the point of causing severe and sometimes life-threatening side effects, referred to as fluoropyrimidine toxicity.

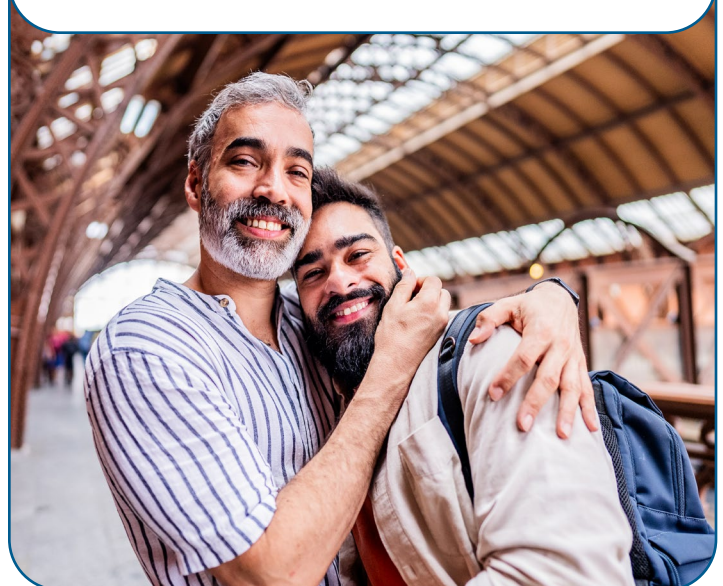
Signs and symptoms often start within several days of the first fluoropyrimidine treatment. Symptoms include mouth sores, abdominal pain, blood in the stool, nausea, vomiting, and diarrhea. The skin on the palms of your

hands and soles of your feet may start peeling and become red, swollen, and painful. Other possible side effects include shortness of breath and hair loss.

If you develop any of these symptoms in the days after receiving capecitabine or 5-FU, contact your care team right away. There is a U.S. FDA-approved treatment for this uncommon side effect of fluoropyrimidines.

If treatment with a fluoropyrimidine is planned, talk to your provider about fluoropyrimidine toxicity. After understanding your risk, you may choose to have genetic testing to learn if you have a gene variant that causes DPD deficiency.

"I learned that getting a second opinion had nothing to do with trusting my doctor. It was the process of getting all the information I needed to make the best decisions for my treatment."



Blood tests

CEA blood test

Carcinoembryonic antigen (CEA) is a protein found in blood. People with colorectal cancer tend to have more of this protein than normal.

Monitoring CEA can be helpful for some cancers that are only in the rectum. If the level rises, it could signal that the cancer has spread. If the level is high at diagnosis, it could suggest that the cancer has already spread. Monitoring CEA isn't helpful for everyone. Pregnant people and tobacco users may also have more CEA than average in their blood.

Complete blood count

Cancer and other health problems can cause low or high blood cell counts. A complete blood count (CBC) measures the number of white blood cells, red blood cells, and platelets in a blood sample.

White blood cells help you fight infection. Red blood cells carry oxygen throughout the body. Platelets help wounds heal by forming blood clots.

Chemistry profile

A chemistry profile (also called a comprehensive metabolic panel) is a group of tests that provides information about how well your kidneys, liver, and other organs are working.



Team approach

When possible, treatment for rectal cancer is best delivered by a team of experts in the following specialties:

- Digestive health (gastroenterology)
- Chemotherapy (medical oncology)
- Surgery (surgical oncology)
- Radiotherapy (radiation oncology)
- Imaging (radiology)

Imaging

Imaging tests can show areas of cancer inside the body. This information helps your care team plan treatment. A radiologist will interpret and convey the imaging results to your oncologist.

Your care team will tell you how to prepare for these tests. You may need to stop taking some medicines and stop eating and drinking for a few hours beforehand.

Some imaging tests use contrast. Contrast is a substance that makes the pictures clearer. It is injected into your vein and/or mixed with a liquid to drink. Tell your doctor if you've had problems with contrast in the past. You may be given or advised to take a steroid and an antihistamine medication (like Benadryl) before the scan. It will lower the risk of an allergic reaction.

MRI

If surgery is needed or being considered, MRI of your pelvis is recommended. MRI can show how deep the cancer has grown into the rectal wall and whether it has spread to nearby lymph nodes.

Special techniques called a staging protocol are used during pelvis MRI to stage the cancer. Contrast may or may not be used. If needed, MRI may also be used to look for cancer spread in the abdomen, especially in the liver.

MRI doesn't use radiation and is safe for most people. Ask your care team what to expect during a pelvis MRI. Tell them if you get nervous in confined spaces. You may be given a sedative (medicine) to help you relax.

Pelvis MRI

MRI can show how deep the cancer has grown into the rectal wall and whether it has spread to nearby lymph nodes. If surgery to remove the rectal tumor is planned or being considered, pelvis MRI is recommended.



2 Testing for rectal cancer

Endorectal ultrasound

In some cases, an endorectal ultrasound (EUS) is performed instead of a pelvis MRI.

Using sound waves, this test can also show the extent of cancer in the rectum and pelvis. A small probe is inserted into the rectum. Echoes form a picture that can be seen by your doctor on a screen. If needed, EUS may also be used to guide a biopsy of lymph nodes or other areas near the rectum.

CT

A CT scan of your chest and abdomen is recommended as part of initial testing. CT can show if the cancer has spread to the liver, lungs, or other organs. Contrast will be used.

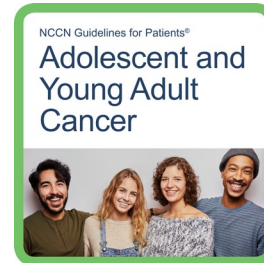
During the scan you will lie face-up on a table that moves through a tunnel-like machine. A technician will be nearby. You will be able to hear and talk to them at all times.

Fertility and family planning

For unknown reasons, colorectal cancer is being diagnosed more often in young adults. Treatment for rectal cancer, especially radiation therapy, can impair your ability to have children.

Radiation can damage the ovaries and testicles, leading to impaired fertility. In people assigned female at birth, damage to the ovaries causes early menopause. Treatment with certain chemotherapies can also impair fertility by damaging sperm or eggs.

See *Chapter 6: Survivorship* for more information on early menopause.



For more information on fertility and family planning, see *NCCN Guidelines for Patients: Adolescent and Young Adult Cancer* at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN NCCN Patient Guides for Cancer](#) app.

If you may want the option of having children after treatment, ask your care team if your planned treatment will affect your fertility. They may refer you to an expert to learn about options for preserving your fertility, including:

- Sperm banking (sperm cryopreservation)
- Egg freezing (oocyte cryopreservation)
- Embryo freezing (embryo cryopreservation)
- Ovarian tissue freezing (cryopreservation)
- Ovarian transposition (oophoropexy)

Key points

- All rectal tumors should be tested for mismatch repair deficiency (dMMR) or microsatellite instability-high (MSI-H). Cancers that don't have either of these biomarkers are described as mismatch repair proficient (pMMR) or microsatellite stable (MSS).
- If the cancer is found to have the dMMR or MSI-H biomarker, genetic testing (of your blood or saliva) for Lynch syndrome is recommended.
- MRI of the pelvis using a special rectal cancer staging technique can show the extent of the cancer in the rectum and nearby lymph nodes.
- CT of the chest and abdomen can show if the cancer has spread to the liver, lungs, abdominal lining, or other organs.
- Testing the tumor for alterations in the PI3K pathway helps guide decisions about taking aspirin after surgery to prevent recurrence.
- After understanding your risk of fluoropyrimidine toxicity, you may choose to be tested for gene variants that cause DPD deficiency.
- If you want the option of having children in the future or are unsure, ask your care team about the fertility-related risks of your treatment plan and options for preserving your fertility.

Questions to ask

- Does the cancer have the dMMR biomarker?
- What if I am pregnant or planning to become pregnant in the future?
- Colorectal cancer runs in my family. What does this mean for my care?
- My doctor said I need fluoropyrimidine chemotherapy. Should *DPYD* testing be performed?
- Is it worth it to go through treatment?

3

Treatment for early rectal cancer

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3 Treatment for early rectal cancer

This chapter explains treatment for polyps with cancer and for small rectal cancers that haven't spread to nearby lymph nodes.

Polyps with cancer

There are 2 main shapes of polyps. Pedunculated polyps grow on a stem and have a rounded top. Sessile polyps are flatter and don't stick out much from the bowel wall.

If cancer has just started to form in a polyp, it's described as a **malignant** (cancerous) **polyp**.

Polypectomy

Most polyps can be removed during a colonoscopy using a minor endoscopic procedure called a polypectomy. In many cases, no further treatment is needed. In other cases, resection (surgery) to remove a larger piece of the rectum is necessary.

Your doctor will consider the factors below to determine whether resection is needed after polypectomy:

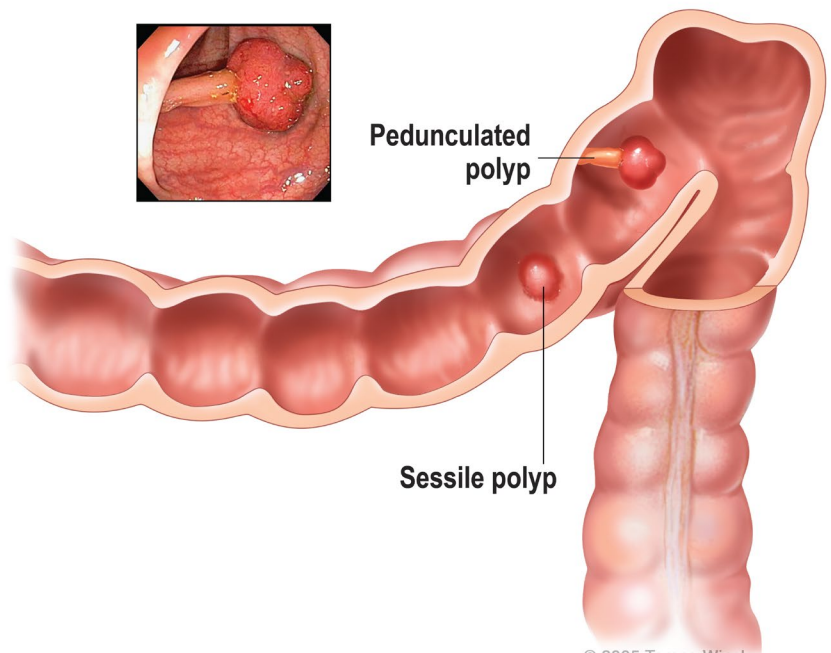
- The size and shape of the polyp
- The polypectomy results
- The results of testing the removed tissue

After a low-risk polyp is removed by polypectomy, surveillance will begin. See below for recommended follow-up testing.

Surveillance after polypectomy

After a polypectomy, follow-up testing is recommended as follows:

- Colonoscopy in 1 year
- Physical exams with proctoscopy every 3 to 6 months for 2 years



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Endoscopic submucosal dissection

Endoscopic submucosal dissection (ESD) is a minimally invasive procedure that removes small rectal cancers in one piece, often curing them.

ESD is a recommended option for **low-risk T1 tumors** as long as nearby lymph nodes appear cancer-free on imaging tests. T1 is the staging term for tumors that haven't grown into the muscle layer of the rectal wall. Low-risk describes tumors without specific features that often lead to recurrence.

First, an endoscope is used to locate the tumor in the rectal wall. A tool is guided through the scope to inject fluid between the tumor and the layer of muscle beneath it. This helps lift and separate the tumor, making it easier to cut off.

Guide 2

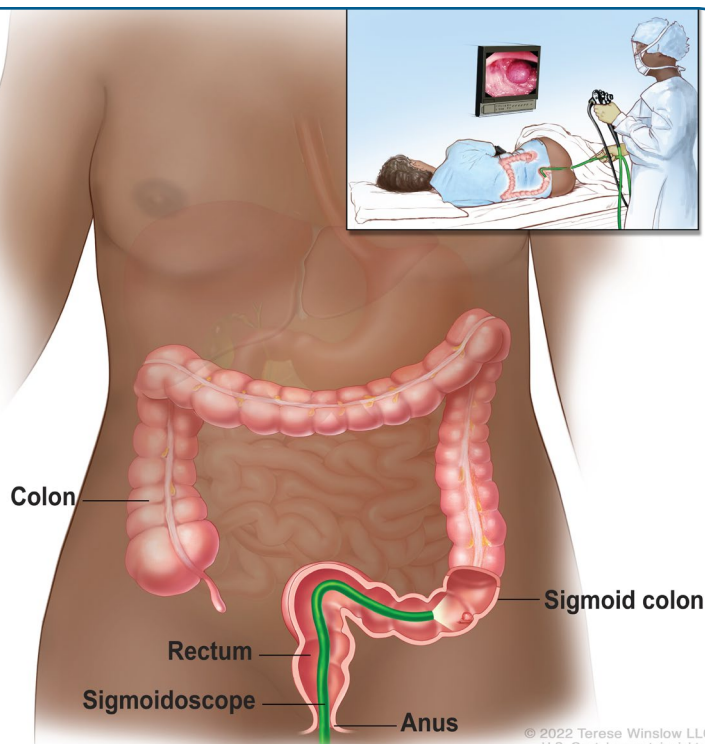
Five-year surveillance after ESD

Flexible sigmoidoscopy	Every 6 months
Endorectal ultrasound or pelvis MRI with contrast	Years 1 and 2: • Every 3 to 6 months Years 3, 4, and 5: • Every 6 months

Most people don't need more treatment. Surveillance with flexible sigmoidoscopy and imaging is recommended for 5 years after ESD. See **Guide 2**.

Flexible sigmoidoscopy

In addition to imaging, flexible sigmoidoscopy is recommended every 6 months for 5 years after ESD. This procedure allows your doctor to check your rectum and sigmoid colon for new cancer growth.



Transanal local excision

If the tumor is located at the end of the rectum, it may be possible to surgically remove it through the anus. This is called transanal local excision. It’s only an option for **low-risk T1 tumors** located close to the anal opening.

The surgeon cuts through all layers of the rectal wall to remove the cancer and some surrounding normal rectal tissue. Lymph nodes aren’t removed.

If surgery confirms that the tumor is T1 and low-risk, no further treatment is needed. Surveillance will begin. See **Guide 3**.



Your medical team is your greatest asset and friend(s), especially the nurses. They are all highly skilled and sincerely dedicated to help you however they can."

Guide 3
Surveillance after transanal local excision

Length of surveillance	5 years
Proctoscopy with endorectal ultrasound or pelvis MRI with contrast	Years 1 and 2: Every 3 to 6 months Years 3, 4, and 5: Every 6 months
Colonoscopy	One year after surgery. The timing of your next colonoscopy will depend on the results.

Abdominal surgery

This type of surgery, also called transabdominal resection, involves cutting into the abdomen to reach and remove the rectal tumor, some surrounding healthy tissue, and nearby lymph nodes. It is the **preferred** treatment for the following rectal tumors:

- T1 tumors with high-risk features
- All T2 tumors

T2 is the staging term for tumors that have invaded the muscle layer of the rectal wall.

Abdominal surgery is also a recommended option for low-risk T1 tumors and some low-risk, larger (T3) tumors that haven't spread to lymph nodes.

The following types of abdominal surgery are described on the next pages:

- Low anterior resection (LAR)
- Abdominoperineal resection (APR)

The type of abdominal surgery performed depends on the location of the tumor and the extent of the cancer. At least 12 lymph nodes should be removed during all types of transabdominal surgery.

The tissue and lymph nodes removed during surgery are sent to a lab for testing. A pathologist assesses how far within the rectal wall the cancer has grown and whether any removed lymph nodes have cancer.

Using this information, they assign the cancer a stage of I (1), II (2), or III (3). This process is called surgical (or pathologic) staging. The

stage is used to determine whether more treatment is needed.

Stage 1 cancer has grown into the second or third (muscle) layer of the rectal wall. No further treatment is needed.

Stage 2 cancer has grown through the rectal wall and possibly into nearby tissues or organs. More treatment is almost always needed.

Stage 3 cancer has spread to nearby lymph nodes. More treatment is almost always needed.

3 Treatment for early rectal cancer

Low anterior resection

A low anterior resection (LAR) is a type of transabdominal surgery used for tumors in the middle or upper rectum.

In addition to the rectal tumor, part or all of the last section of the colon, called the sigmoid colon, is removed. When possible, the colon is connected to the remaining rectum. This is called a colorectal anastomosis (reconnection). This allows for the possibility of having near-normal bowel movements.

If nearly all of the rectum must be removed, the colon may be attached directly to the anus. This is called a coloanal anastomosis.

Your surgeon is likely to delay reattaching the bowel to let the surgical site heal. In this case, an ileostomy is created. This connects the end of the small intestine (ileum) directly

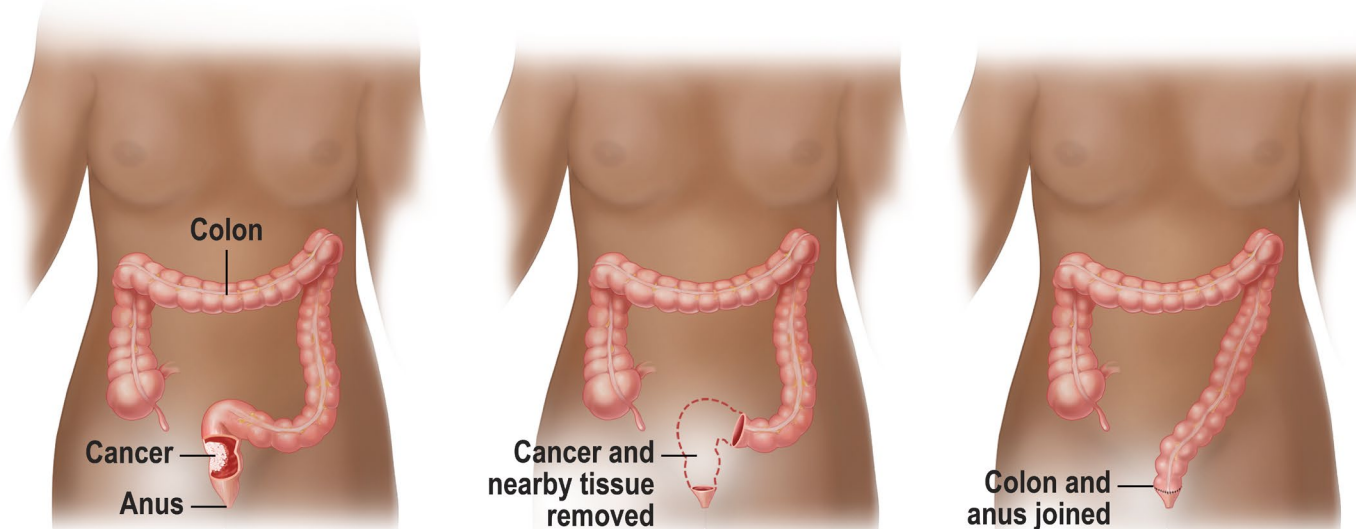
to the surface of your abdominal wall through an opening called a stoma. An ostomy bag is attached with adhesive to the skin around the stoma to capture stool.

In some cases, a colostomy is made instead of an ileostomy. A colostomy connects a part of the upper colon directly to the surface of the abdominal wall and allows stool to exit the body into a colostomy bag.

An ileostomy or colostomy may only be needed for several months after LAR, to allow the rectum to heal before being reconnected to the colon. But sometimes it is needed permanently.

LAR can cause a group of symptoms called low anterior resection syndrome (LARS). See *Chapter 6: Survivorship* for information on managing LARS.

Low anterior resection (LAR) is a type of abdominal surgery used to remove tumors in the middle or upper rectum. The rectal tumor and part or all of the sigmoid colon is removed. When possible, the colon is connected to the remaining rectum.



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Abdominoperineal resection

This type of transabdominal surgery is used for tumors in the lower rectum. These tumors may have grown into the anus or nearby pelvic floor muscle.

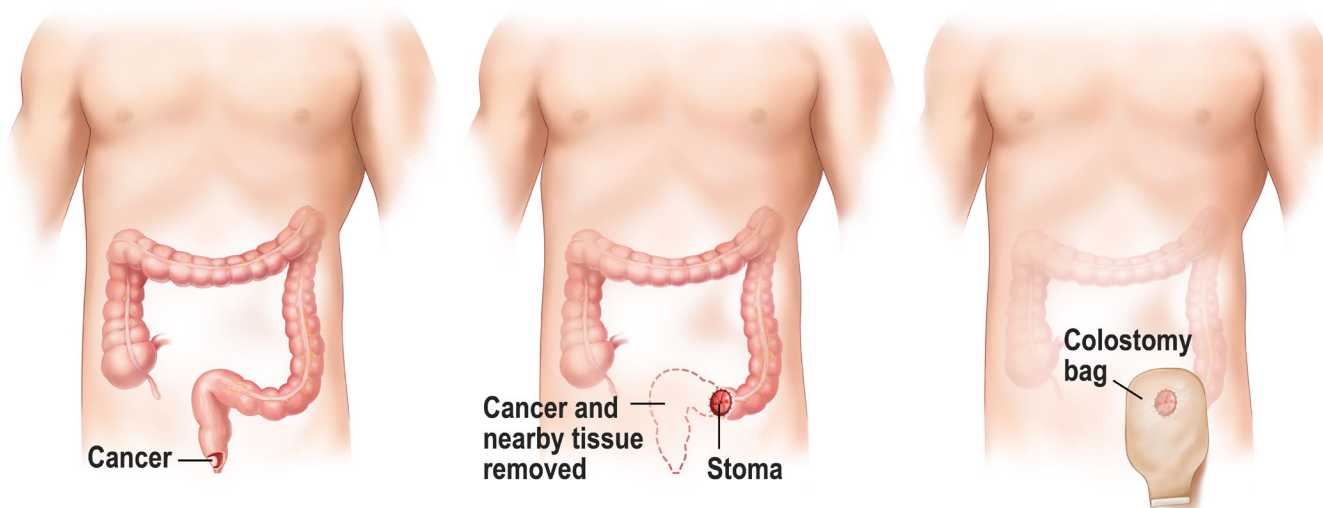
Abdominoperineal resection (APR) typically involves removing the anus, sphincter muscles, rectum, the area where the rectum and colon meet, and other surrounding tissue.

A permanent colostomy is always needed after APR surgery. Consider joining an ostomy support group or seeing a health care provider that specializes in ostomy care. See *Chapter 6: Survivorship* for more information.

“

Be your own advocate. Talk to people who have gone through the same thing. Ask a lot of questions, even the ones you are afraid to ask.”

Abdominoperineal resection (APR) is used for tumors in the lower rectum that may have grown into the anus or nearby muscle. It involves removing the rectum and anus and creating a permanent colostomy for stool to leave the body.



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After abdominal surgery

Sometimes abdominal surgery to remove a small (T1 or T2) tumor finds that the cancer is larger or deeper than expected, or that nearby lymph nodes contain microscopic cancer cells. As a result, the cancer stage may be upgraded to stage 2 or 3. In this case, chemotherapy, with or without radiation therapy, often comes next. In very specific cases, observation without treatment may be an option.

The order in which you receive these treatments can vary. Your care team will determine the best sequence for you. More information on these treatments is provided next.

Chemotherapy

Chemotherapy given after surgery to kill remaining cancer cells is called adjuvant chemotherapy. Several months of either the FOLFOX or CAPEOX regimen is recommended.

FOLFOX includes the following medicines:

- Folinic acid (leucovorin calcium)
- Fluorouracil (5-FU)
- Oxaliplatin

CAPEOX includes:

- Capecitabine
- Oxaliplatin

Chemotherapy is given in cycles that include some treatment days followed by days of rest. This allows your body to recover between cycles. Cycles vary in length depending on which drugs are used.

ctDNA Testing

After surgery to remove a colon or rectal tumor, microscopic pieces of DNA released by the tumor may remain in your bloodstream. If ctDNA is detected in the blood after surgery, the cancer is likely to return at some point. **But, cancer experts don't currently recommend ctDNA testing to guide decisions about your care.**

Your care team will explain the possible side effects of chemotherapy. Importantly, oxaliplatin can cause nerve damage (called neuropathy) to your fingers and toes. Symptoms include numbness, cramping, tingling, or pain in these areas. Neuropathy can be permanent.

Long-course chemoradiation

Chemoradiation is used most often to treat locally advanced cancers. For details, see *Chapter 4: Treatment for locally advanced rectal cancer*.

3 Treatment for early rectal cancer

Surveillance

For cancers removed by abdominal surgery, recommended surveillance testing is shown in **Guide 4**.

In addition to surveillance testing, a range of other care is important for cancer survivors. See *Chapter 6: Survivorship* for more information.

Aspirin

For stage 2 and 3 cancers with alterations that activate PI3K, taking aspirin for 3 years after surgery is recommended to lower the risk of recurrence. You won't start taking it until you have recovered from surgery.

Guide 4 Surveillance after abdominal surgery		
Stage 1 (determined by surgical staging)	Colonoscopy	One year after surgery. The timing of your next colonoscopy will depend on the results. Other testing is only needed if you develop symptoms or if your provider suspects recurrence.
	Physical exam and CEA blood test	Years 1 and 2: Every 3 to 6 months Years 3, 4, and 5: Every 6 months
Stages 2 and 3 (determined by surgical staging)	CT of your chest, abdomen, and pelvis	Every 6 to 12 months for 5 years
	Colonoscopy	1 year after surgery if you had a complete colonoscopy at diagnosis 3 to 6 months after surgery if you didn't have a complete colonoscopy at diagnosis

Key points

- Polypectomy may be the only treatment needed for a polyp that contains cancer. In other cases, surgery may be needed to remove a larger piece of the rectum.
- Surgery is recommended for early rectal cancers when possible. T1 is the staging term for tumors that haven't grown into the muscle layer of the rectal wall. T2 tumors have invaded the muscle layer.
- Options for low-risk T1 tumors include endoscopic submucosal dissection (ESD) and transanal local excision. ESD removes small rectal cancers in one piece, often curing them. Transanal local excision removes small tumors located close to the anal opening.
- Transabdominal resection is preferred for high-risk T1 tumors and all T2 tumors. Types of transabdominal resection include low anterior resection (LAR) and abdominoperineal resection (APR).
- For small tumors found to be a more advanced stage after abdominal surgery, treatment with chemotherapy with or without long-course chemoradiation is often recommended.
- For stage 2 and 3 cancers with alterations in the PI3K gene pathway, taking aspirin to help prevent recurrence is recommended for 3 years after abdominal surgery.

Questions to ask

- Do I need surgery, and if so what type?
- Will I need an ostomy and, if so, will it be permanent or temporary?
- When treatment is over, how often should I get checked for new polyps or cancer regrowth?
- Is recurrence likely after treatment for small rectal tumors?



**Let us know what
you think!**

**Please take a moment to
complete an online survey about
the NCCN Guidelines for Patients.
[NCCN.org/patients/response](https://www.nccn.org/patients/response)**

4

Treatment for locally advanced rectal cancer

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- 28 Radiation therapy
 - 29 Long-course chemoradiation
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- 30 Treatment for pMMR or MSS cancer
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- 32 Recurrence
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This chapter describes treatment for rectal cancer that has grown beyond the rectal wall but hasn't spread to organs outside of the inner pelvis. Treatment depends on the mismatch repair (MMR) status of the cancer.

What is locally advanced rectal cancer?

Rectal cancer is described as locally advanced if any one of the following is true:

- It has grown through the rectal wall (a T3 or T4 tumor)
- It has spread to nearby lymph nodes
- It is confined to the pelvis but can't be surgically removed.

Radiation therapy

Treatment for locally advanced rectal cancer often involves external beam radiation therapy (EBRT), chemotherapy, or both.

In EBRT, radiation is delivered using a large machine located outside the body. The radiation beams pass through skin and other tissue to reach the tumor and surrounding tissue. The radiation beams are shaped to the cancer site. This helps minimize damage to healthy tissue.

Side effects of radiation therapy

- Feeling tired and worn out
- Hair loss in the treated area
- Changes to urination and bowel movements
- Diarrhea
- Nausea or vomiting
- Late side effects can include infertility, sexual problems, bowel problems, reduced bone density, and second cancers
- Vaginal dilators can help with the sexual side effects of radiation therapy – see *Chapter 6: Survivorship* for more information
- Ask your care team for a full list of possible side effects

There are different types of EBRT. The type used depends on the location and size of the tumor and other factors.

Long-course chemoradiation

EBRT is often used with chemotherapy to treat rectal cancer. When given with radiation, chemotherapy can make radiation better at killing cancer cells. This is called radiosensitizing chemotherapy.

In long-course chemoradiation, radiation therapy is given in 25 to 28 treatment sessions, called fractions. Most people have 5 sessions per week. Chemotherapy is given on radiation days during the same 5- to 6-week period.

Either 5-fluorouracil (5-FU) or capecitabine is usually given for chemotherapy. 5-FU is given using a continuous infusion pump hooked into a central access device like a port-a-cath.

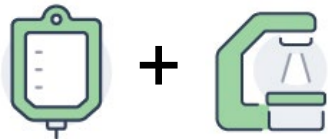
You are connected to this pump at the start of radiation and wear it everywhere. The pump is refilled with 5-FU every 5 to 7 days until radiation therapy is complete.

Capecitabine is a pill that you take twice daily on radiation days.

Short-course radiation therapy

Another method of radiation treatment for rectal cancer is short-course radiation therapy. This method delivers a higher dose of radiation per treatment session in comparison to long-course radiation, and is typically complete in only 5 treatment sessions. Chemotherapy isn't given.

Long-course chemoradiation



- 25 to 28 radiation treatment sessions (Monday through Friday)
- Chemotherapy is given during the same time period
- Either 5-fluorouracil or capecitabine is usually given for chemotherapy

Short-course radiation therapy



- High-dose radiation is given over a short time period
- Typically only 5 treatment sessions are needed

For many years, long-course chemoradiation was the standard approach for treating locally advanced rectal cancer. Short-course radiation therapy is another option found to have similar outcomes in certain cases.

Treatment for pMMR or MSS cancer

Treatment for locally advanced mismatch repair proficient (pMMR) or microsatellite stable (MSS) cancers typically begins with 12 to 16 weeks of chemotherapy and either long-course chemoradiation or short-course radiation therapy. For most people, transabdominal surgery is the next step. This approach is known as **total neoadjuvant therapy (TNT)**.

Sometimes chemotherapy and chemoradiation work so well that no cancer can be detected in the body. This is called a complete response. If this happens, research suggests that

having surgery doesn't always lead to better outcomes.

You may be given the option to monitor the cancer closely instead. This is only an option for carefully selected patients who agree to close surveillance programs. See **Guide 5**.

The benefits and risks of taking a watch-and-wait approach instead of having surgery aren't fully known. The decision to have surgery (or not) should be one that you make with your doctor after careful consideration.

Guide 5
Surveillance after non-surgical treatment

Physical exam and carcinoembryonic antigen (CEA) blood test	The first 2 years: Every 3 to 6 months The following 3 years: Every 6 months
Digital rectal exam (DRE) and proctoscopy or flexible sigmoidoscopy	The first 2 years: Every 3 to 4 months The following 3 years: Every 6 months
MRI of your rectum	Every 6 months for up to 3 years
CT of your chest and abdomen	Every 6 to 12 months for 5 years. When MRI imaging stops after 3 years, CT scanning will also include your pelvis.
Colonoscopy	In 1 year. The timing of your next colonoscopy will depend on the results.

Treatment for dMMR or MSI-H cancer

The 2 main treatment approaches for locally advanced mismatch repair deficient (dMMR) or microsatellite instability-high (MSI-H) cancers are described next.

Guide 6 Checkpoint inhibitors

Generic name	Brand name
Atezolizumab	Tecentriq
Avelumab	Bavencio
Cemiplimab-rwlc	Libtayo
Cosibelimab-ipdl	Unloxcyt
Dostarlimab-gxly	Jemperli
Durvalumab	Imfinzi
Ipilimumab + Nivolumab	Opdivo + Yervoy
Nivolumab	Opdivo
Pembrolizumab	Keytruda
Penpulimab-kcqx	Anikora
Retifanlimab-dlwr	Zynyz
Tislelizumab-jsgr	Tevimbra
Toripalimab-tpzi	Loqtorzi

Immunotherapy

The **preferred** approach for locally advanced dMMR or MSI-H cancers begins with immunotherapy with a checkpoint inhibitor. Immunotherapy boosts your immune system to help it better find and kill cancer cells. Currently recommended checkpoint inhibitors are listed at left in **Guide 6**.

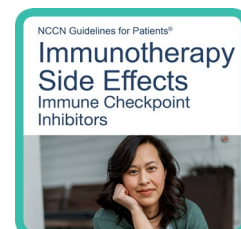
Immunotherapy is recommended for up to 6 months. If it works so well that no cancer remains, treatment is finished. You will begin surveillance to find recurrence early. For surveillance after non-surgical treatment, see **Guide 5** on page 30.

If cancer remains after 6 months of immunotherapy, treatment with one of the following is recommended:

- Long-course chemoradiation
- Short-course radiation therapy

Most people have transabdominal surgery next. Or, if no cancer can be detected in the body, it's called a complete response. If you have a **complete response**, observing the cancer without treatment may be an option. You will have close surveillance to check for recurrence. See **Guide 5**.

Checkpoint inhibitors have unique side effects. Learning to recognize possible reactions can help you report them early to your care team. More information is available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



Total neoadjuvant therapy

If you can't have immunotherapy, another recommended approach is called total neoadjuvant therapy. Treatment begins with either long-course chemoradiation or short-course radiation therapy. The next step is 12 to 16 weeks of chemotherapy, usually with FOLFOX or CAPEOX. Your provider may also consider the FOLFIRINOX regimen.

After treatment, if imaging results show cancer, surgery is recommended. If you aren't a candidate for surgery, chemotherapy is an option.

See **Guide 4** on page 25 for recommended surveillance after abdominal surgery.

Recurrence

If cancer returns to the rectum or to most areas in the pelvis, it's called a **local recurrence**.

Your options for treatment will depend on many factors including:

- The number and location of recurrent tumor(s)
- Treatments you've already received
- Whether the tumor has any biomarkers
- How well you're expected to tolerate certain treatments

Local recurrences should be evaluated by a multidisciplinary care team to determine the best treatment approach for you.

When cancer returns to areas far from the rectum, such as the liver or lungs, it's called a **distant or metastatic recurrence**. See *Chapter 5: Treatment for metastatic rectal cancer* for treatment information.

Key points

- Locally advanced cancers have grown through the rectal wall or have spread to nearby lymph nodes, but haven't spread outside of the pelvis.
- Treatment for locally advanced rectal cancer often involves external beam radiation therapy (EBRT), chemotherapy, or both.
- In long-course chemoradiation, radiation therapy is given in 25 to 28 treatment sessions. Chemotherapy is given during the same 5- to 6-week period.
- Short-course radiation therapy delivers a higher dose of radiation per treatment session than long-course radiation, and is typically complete in 5 treatment sessions.
- Treatment for locally advanced mismatch repair proficient (pMMR) or microsatellite stable (MSS) cancers starts with chemotherapy and either long-course chemoradiation or short-course radiation therapy. Surgery usually follows. This approach is known as total neoadjuvant therapy.
- The preferred treatment for locally advanced mismatch repair deficient (dMMR) or microsatellite instability-high (MSI-H) cancers is 6 months of immunotherapy. If needed, long-course chemoradiation or short-course radiation therapy is given afterward. Transabdominal surgery is often the next step.
- If you can't have immunotherapy, a second option for locally advanced dMMR cancers begins with either chemoradiation or radiation therapy, followed by

chemotherapy alone. Surgery will follow if needed and possible.

Questions to ask

- Will I have total neoadjuvant therapy?
- What is the simulation session for radiation therapy? What can I expect?
- How likely is it that surgery will be possible after total neoadjuvant therapy?
- What follow-up testing is needed for my cancer stage?
- How likely is the cancer to return after treatment? Can I do anything to lower the risk?

5

Treatment for metastatic rectal cancer

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Planning treatment for metastatic rectal cancer starts with biomarker testing.

If not found and treated early enough, cancer cells will break off from the rectal tumor and travel in lymph or blood to other areas of the body—usually the liver, lungs, or abdomen. This process is called metastasis. Treatment for metastatic rectal cancer depends on the results of biomarker testing, described next.

Biomarker testing

Biomarkers are features of a cancer that can help guide your care. Many biomarkers are mutations (changes) in specific genes.

When possible, biomarker testing is performed on a piece of tumor tissue removed during a biopsy or surgery. Or, if needed, a sample of your blood can be tested instead.

All metastatic rectal cancers should be tested for the following biomarkers:

- *KRAS* and *NRAS* mutations
- *BRAF* V600E mutations
- HER2 amplification
- Mismatch repair (MMR) or microsatellite instability (MSI) status, if unknown

Testing for many biomarkers at one time is called multigene panel testing (MGPT). MGPT can find other rare biomarkers for which targeted therapies are available, such as *POLE/POLD1* mutations, *RET* gene fusions, and *NTRK* gene fusions.

Local therapies

Your options for treating liver and lung tumors may include local therapies. These are treatments that remove or destroy tumors without damaging the surrounding area.

Some are interventional oncology/radiology techniques, also known as image-guided therapies. These techniques use imaging, such as ultrasound, CT, or PET, to deliver minimally invasive cancer treatments. Using imaging during the procedure allows your doctor to precisely target the tumor(s) and confirm treatment success.

To learn if treatment with surgery or other local therapies is an option, your case should be reviewed by a multidisciplinary team of experts. This team should include a surgeon experienced in removing liver and lung tumors and an interventional oncologist/radiologist.

Resection

Surgery to remove liver or lung tumors is called resection. Resection may be paired with ablation (described next) to completely remove small tumors.

If a liver resection is needed, your liver may need to be enlarged first. This is done using a minimally invasive procedure called portal vein embolization.

Tumor ablation

Image-guided ablation is a type of treatment that destroys small liver or lung tumors with minimal damage to surrounding normal tissue. It may be paired with surgery to completely remove a tumor or used alone for small tumors that can be fully removed or destroyed.

5 Treatment for metastatic rectal cancer

Thermal ablation methods use extreme heat (most common) or extreme cold to destroy tumors. Either an interventional radiologist or a surgeon places a special needle called a probe into or next to the tumor. The probe delivers energy to the tumor while minimizing damage to surrounding normal tissue.

A non-thermal method that uses electrical pulses may be helpful for tumors located in areas where resection or thermal ablation can't be used.

Intra-arterial therapies

If the cancer has spread mainly to the liver and other treatments haven't worked or can't be used, intra-arterial liver-directed therapies may be an option. These therapies treat liver tumors with:

- Chemotherapy beads (chemoembolization)
- Radioactive spheres (radioembolization)

Use of radioactive spheres is known as selective internal radiation therapy (SIRT) or transarterial hepatic radioembolization (TARE). These procedures are performed by interventional oncologists/radiologists.

In the rare cases where chemotherapy and radiation can't be used, beads that don't contain chemotherapy may be used to physically block blood supply to the tumor. This is called bland embolization.

HAIC

Hepatic arterial infusion chemotherapy (HAIC) is chemotherapy given directly to the liver. Through use of a port or pump that is usually placed during surgery, the drugs are funneled directly into the artery leading to the liver.

HAIC should only be performed by medical oncologists with experience in this method.

Stereotactic body radiation therapy

Stereotactic body radiation therapy (SBRT) may be used to treat rectal cancer that has spread to the liver, lungs, or lymph nodes.

SBRT delivers high doses of radiation to metastatic tumor(s) using very precise beams. The radiation comes from a large machine outside the body. The radiation passes through skin and other tissue to reach the tumor(s). Treatment with SBRT is typically complete in 5 or fewer sessions, called fractions.

Stage 4 pMMR or MSS cancer in the liver or lungs

Treatment for stage 4 rectal cancer depends on whether the cancer is resectable.

The cancer is resectable if the rectal tumor can be safely resected and the liver or lung tumors can be completely removed or destroyed using resection and other local therapies.

But even resectable cancers need other treatment before you can have surgery.

Resectable cancer

The preferred approach for resectable mismatch repair proficient (pMMR) or microsatellite stable (MSS) cancers begins with chemotherapy. FOLFOX and CAPEOX are preferred options (see below to learn what makes a systemic therapy **preferred**).

After chemotherapy, most people receive short-course radiation therapy or chemoradiation, followed by surgery to remove the rectal tumor. The liver and lung tumors may be resected or destroyed at this time, or at a later date.

While the approach just described is preferred, another option is to receive short-course radiation or chemoradiation first, followed by chemotherapy, and then surgery.

Regardless of the order in which they are given, sometimes chemotherapy and radiation therapy work so well that surgery isn't needed. In this case, surveillance will begin. See **Guide 5**.

SBRT for metastatic tumors

A highly specialized type of radiation therapy called SBRT may be used to treat rectal cancer that has spread to the liver, lungs, or bone. Treatment typically takes 5 or fewer sessions.



5 Treatment for metastatic rectal cancer

What is a preferred therapy?

All systemic treatments listed in this guide are recommended and appropriate. When helpful, NCCN experts also assign a level of preference to their recommendations for systemic therapies:

- **Preferred therapies** have the most evidence they may work better and may be safer than other therapies.
- **Other recommended therapies** can provide effective results but may have less evidence, more side effects, or may not work quite as well as preferred therapies.
- **Therapies used in certain cases** work best for individuals with specific cancer features or health circumstances.

Unresectable cancer

Unresectable stage 4 pMMR or MSS cancers are treated with chemotherapy. One of the following regimens is recommended first (as first-line therapy). A treatment called bevacizumab (Avastin) may be added to any of these regimens:

- FOLFIRI
- FOLFOX
- CAPEOX
- FOLFIRINOX

Depending on the results of biomarker testing, you may be eligible to receive an EGFR blocker in addition to chemotherapy. In cancers without certain mutations, pairing these medicines often has very good treatment results. Recommended EGFR blockers include cetuximab (Erbix) and panitumumab (Vectibix).

Unresectable stage 4 cancer doesn't typically become resectable during first-line chemotherapy. Most people continue systemic therapy to slow the growth and spread of the cancer. Local therapies may also be an option for controlling liver and lung tumors.

If the cancer progresses (worsens) during recommended first-line chemotherapy, there are good options for second- and even third-line treatment. Your specific options are based on which medicines you've already had. And there are also less intensive regimens that can be used for any line of treatment.

If the cancer progresses through all recommended regimens, the following may be options:

- Fruquintinib (Fruzaqla)
- Trifluridine and tipiracil (Lonsurf) (bevacizumab may be added)
- Regorafenib (Stivarga)

These therapies are tablets that can be taken by mouth.

Stage 4 dMMR or MSI-H cancer in the liver or lungs

Treatment for stage 4 dMMR or MSI-H cancers starts with immunotherapy. Currently recommended checkpoint inhibitors are listed in **Guide 6**.

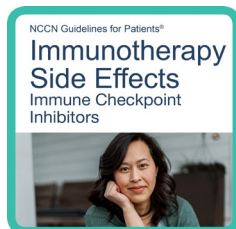
If no cancer can be detected in the body after immunotherapy, it's called a complete response. You will start surveillance testing. See **Guide 5**.

Or, if imaging tests show cancer in the body after immunotherapy, surgery is recommended when possible. The liver and lung tumors may be resected or destroyed when the rectal tumor is resected, or at a later date. Your oncologist may recommend either short-course radiation therapy (preferred) or chemoradiation before surgery.

If you've already received checkpoint inhibitor immunotherapy, you aren't a candidate for most of the checkpoint inhibitors in **Guide 6**. Recommended options for treatment include:

- Ipilimumab + Nivolumab (Opdivo + Yervoy)
- Oxaliplatin-based chemotherapy

Checkpoint inhibitors have unique side effects. Learning to recognize possible reactions can help you report them early to your care team. More information is available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



Stage 4 cancer in the abdomen

Some people with metastatic rectal cancer also form tumors in the thin layer of tissue that lines the abdomen, called the peritoneum. The peritoneum covers most of the organs in the abdomen.

Most people receive treatment with systemic therapy. The goal is to relieve or prevent symptoms. Your care team will select a regimen based on whether the tumor has any biomarkers and your overall health.

If the tumor is blocking stool from moving and leaving the body, the bowel needs to be unblocked before systemic therapy is started. This is done using a surgical technique or with a mesh metal tube called a stent.

Surveillance

If you have a complete response to treatment that included surgery, the recommended surveillance schedule is shown in **Guide 7**.

In addition to surveillance testing, a range of other care is important for cancer survivors. For more information, see *Chapter 6: Survivorship*.

Distant recurrence

After treatment for non-metastatic rectal cancer, the cancer may return and spread to the liver, lungs, or other areas. This is called a distant recurrence.

Treatment with surgery, ablation, or other local therapies is recommended if all of the tumors can be totally removed. But this is rare, and most people receive treatment with systemic therapy.

While uncommon, systemic therapy may shrink the tumors enough to be removed with surgery or ablation. If the cancer doesn't become small enough to be resected or ablated, systemic therapy is typically continued. The goal is to slow the growth and spread of the cancer.

Treatment recommendations are provided next according to the mismatch repair status of the cancer and whether resection is possible.

Guide 7

Surveillance after surgical treatment for stage 4 rectal cancer

- **Physical exam**
- **CEA blood test**
- **CT scan of your chest, abdomen, and pelvis**

First 2 years: Every 3 to 6 months

Next 3 years: Every 6 months

Colonoscopy

Did you have a complete colonoscopy at diagnosis?

If you **didn't**, your next colonoscopy is recommended in 3 to 6 months.

If you **did**, you can wait until a year after surgery for your next colonoscopy.

5 Treatment for metastatic rectal cancer

Unresectable pMMR or MSS cancer

If you've had recent treatment with FOLFOX or CAPEOX, you shouldn't have more chemotherapy that includes oxaliplatin. Oxaliplatin can cause serious nerve damage. Your options for systemic therapy will depend, in part, on any biomarkers found in the cancer. See **Guide 8**.

Resectable pMMR or MSS cancer

For resectable pMMR or MSS cancers, treatment typically involves chemotherapy in addition to resection and treatment with local therapies. Chemotherapy may be given before surgery, after surgery, or both.

Guide 8

Distant recurrence of unresectable pMMR/MSS cancer

Biomarkers (features) found in the cancer	Options for initial systemic therapy
<i>BRAF</i> V600E mutation	A <i>BRAF</i> inhibitor called encorafenib (Braftovi) + an EGFR blocker – chemotherapy with FOLFIRI, FOLFOX, or CAPEOX may be added
HER2 overexpression	Your options may include: • Antibody drug conjugate therapy with fam-trastuzumab deruxtecan-nxki (Enhertu) • HER2 blocker therapy + trastuzumab (Herceptin) – for cancers with normal <i>KRAS</i>, <i>NRAS</i>, and <i>BRAF</i> genes. Recommended HER2 blockers include pertuzumab (Perjeta) and tucatinb (Tukysa).
<i>KRAS</i> G12C mutation	A <i>KRAS</i> inhibitor + an EGFR blocker • Adagrasib (Krazati) and sotorasib (Lumakras) are recommended <i>KRAS</i> inhibitors
All other tumors	Chemotherapy with FOLFIRI or irinotecan – depending on the cancer's biomarker profile, either an EGFR blocker or an angiogenesis inhibitor may be given with chemotherapy.

Recommended EGFR blockers include cetuximab (Erbix) and panitumumab (Vectibix)
Recommended angiogenesis inhibitors include bevacizumab (Avastin – preferred), ziv-aflibercept (Zaltrap), and ramucirumab (Cyramza)

5 Treatment for metastatic rectal cancer

If you **haven't** had any chemotherapy, treatment with resection and local therapies is often given first, followed by chemotherapy.

Unresectable dMMR or MSI-H cancer

If you are a candidate and haven't had immunotherapy, treatment with a checkpoint inhibitor is recommended. Checkpoint inhibitors are listed on page 31 in **Guide 6**.

Your doctor will check the extent of the cancer every 2 to 3 months. Surgery may become possible. Or you may continue immunotherapy or switch to a different systemic therapy.

If you've already received checkpoint inhibitor immunotherapy, you aren't a candidate for most of the recommended checkpoint inhibitors. Your options for treatment include:

- Ipilimumab + Nivolumab (Opdivo + Yervoy)
- Oxaliplatin-based chemotherapy

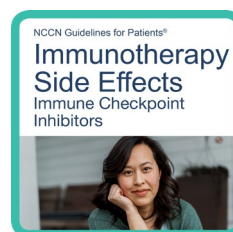
Resectable dMMR or MSI-H cancer

If you are a candidate and haven't had immunotherapy, checkpoint inhibitor immunotherapy is the **preferred** treatment for resectable mismatch repair deficient (dMMR) or microsatellite instability-high (MSI-H) cancer. Recommended checkpoint inhibitors are listed on page 31 in **Guide 6**.

If you have a complete response to immunotherapy, surveillance will resume. Or, if cancer remains, it will be resected or destroyed using local therapies.

Immunotherapy side effects

Checkpoint inhibitors have unique side effects. Unlike other cancer treatments, the side effects of immunotherapy occur because the immune system is attacking healthy cells in the body. Learning to recognize possible reactions can help you report them sooner to your care team. More information is available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



Instead of immunotherapy, having surgery (or treatment with other local therapies) first is also a recommended option, followed by chemotherapy.

If you've had prior immunotherapy, one recommended option is resection first, followed by either chemotherapy or observation. A second option is 2 to 3 months of chemotherapy first, followed by resection. More chemotherapy may follow.

Supportive care

Supportive care is an important part of cancer care. The goal is to improve your quality of life during and after cancer treatment. Supportive care is for everyone with cancer and their families, not just for those at the end of life.

Supportive care includes a wide range of services. Supportive care prevents or manages the symptoms of cancer and the side effects of cancer treatment, like pain and cancer-related fatigue. It also addresses the mental, social, emotional, and spiritual concerns faced by people with cancer. Supportive care provides help with additional needs, such as:

- Making treatment decisions
- Coordinating your care
- Paying for care
- Planning for advanced care and end of life

Read more about the types of support you may receive in *NCCN Guidelines for Patients: Palliative Care*, available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app.



Clinical trials

A clinical trial is a type of medical research study. After being developed and tested in a lab, potential new ways of fighting cancer need to be studied in people.

If found to be safe and effective in a clinical trial, a drug, device, or treatment approach may be approved by the U.S. FDA.

Everyone with cancer should carefully consider all of the treatment options available for their cancer type, including standard treatments and clinical trials. Talk to your doctor about whether a clinical trial may make sense for you.

Phases

Most cancer clinical trials focus on treatment and are done in phases.

- **Phase 1** trials study the safety and side effects of an investigational drug or treatment approach.
- **Phase 2** trials study how well the drug or approach works against a specific type of cancer.
- **Phase 3** trials test the drug or approach against a standard treatment. If the results are good, it may be approved by the FDA.
- **Phase 4** trials study the safety and benefit of an FDA-approved treatment.

Who can enroll?

It depends on the clinical trial's rules, called eligibility criteria. The rules may be about age, cancer type and stage, treatment history, or general health. They ensure that participants are alike in specific ways and that the trial is as safe as possible for the participants.

Informed consent

Clinical trials are managed by a research team. This group of experts will review the study with you in detail, including its purpose and the risks and benefits of joining. All of this information is also provided in an informed consent form. Read the form carefully and ask questions before signing it. Take time to discuss it with people you trust. Keep in mind that you can leave and seek treatment outside of the clinical trial at any time.

Will I get a placebo?

Placebos (inactive versions of real medicines) are almost never used alone in cancer clinical trials. It is common to receive either a placebo with a standard treatment, or a new drug with a standard treatment. You will be informed, verbally and in writing, if a placebo is part of a clinical trial before you enroll.

Are clinical trials free?

There is no fee to enroll in a clinical trial. The study sponsor pays for research-related costs, including the study drug. But you may need to pay for other services, like transportation or childcare, due to extra appointments. During the trial, you will continue to receive standard cancer care. This care is often covered by insurance.



Finding a clinical trial

In the United States

NCCN Cancer Centers
[NCCN.org/cancercenters](https://www.nccn.org/cancercenters)

The National Cancer Institute (NCI)
[cancer.gov/about-cancer/treatment/clinical-trials/search](https://www.cancer.gov/about-cancer/treatment/clinical-trials/search)

Worldwide

The U.S. National Library of Medicine (NLM)
clinicaltrials.gov

Need help finding a clinical trial?

NCI's Cancer Information Service (CIS)
1.800.4.CANCER (1.800.422.6237)
[cancer.gov/contact](https://www.cancer.gov/contact)

Key points

- All metastatic rectal cancers should be tested for biomarkers, including *KRAS*, *NRAS*, and *BRAF* V600E mutations, HER2 amplification, and mismatch repair deficiency (if not already performed).
- Multigene panel testing (MGPT) can find other, rare biomarkers for which targeted treatments may be available.
- Treatment options for metastatic rectal cancer may include local therapies, such as image-guided thermal ablation and stereotactic body radiation therapy (SBRT).
- Stage 4 cancer that can't be removed with surgery and local therapies is treated with systemic therapy.
- For those who are eligible, checkpoint inhibitor immunotherapy is recommended for mismatch repair deficient (dMMR) and POLE/POLD1-mutated metastatic cancers.
- Supportive care is available to help with your symptoms, make you more comfortable, and improve your quality of life.
- Clinical trials provide access to investigational treatments that may, in time, be approved by the FDA.

Questions to ask

- Am I a candidate for a clinical trial? Do you know of one I can join?
- Am I a candidate for surgery? If not, is it possible that I'll become a candidate?
- Does my cancer have any biomarkers? How does this affect my options?
- Did an interventional oncologist/radiologist review my case, along with a multidisciplinary team, to see if I am a candidate for local therapies?
- Which systemic therapy regimen do you recommend for me? Why?

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Survivorship

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Survivorship begins on the day you learn you have rectal cancer. It focuses on the physical, emotional, and financial issues unique to cancer survivors.

Rectal cancer survivors may experience both short- and long-term health effects of cancer and its treatment. The effects depend in part on the treatment(s) received. Surgery, anti-cancer medicines, and radiation therapy all have unique potential side effects.

Bowel changes

Surgery that involves removing all or part of the rectum is called a low anterior resection (LAR). After this surgery, it's common to experience changes in your bowel movements and habits. These symptoms, listed below, are referred to as **low anterior resection syndrome (LARS)**:

- Increase in bowel movements
- Urgent need to pass stool
- Having many (possibly watery) bowel movements over the course of a few hours (stool clustering)
- Irregular and unpredictable bowel movements
- Inability to control bowel movements (fecal incontinence)
- Increase in gas (flatulence)

Loperamide (Imodium) is an over-the-counter anti-diarrheal medicine. It can help with stool clustering and incontinence. Exercises that strengthen the muscles in the pelvic floor can help with urgency and stool incontinence.

Making changes to your diet can also help with LARS symptoms. Eat foods high in insoluble (not soluble) fiber and use fiber supplements. Psyllium-based products (like Metamucil) may help slow and thicken the stool.

Ask your care team for a list of foods that may help or worsen bowel changes after low anterior resection. Many people find that protective undergarments provide peace of mind. A cancer nutritionist can also help you make changes to your diet after rectal surgery.

Nerve damage

The chemotherapy drug oxaliplatin can damage nerves in your fingers and toes. Symptoms include numbness, cramping, tingling, or pain in these areas. Acupuncture and heat may help.

If you have painful nerve damage, a drug called duloxetine (Cymbalta) may help. If the pain is persistent, talk to your doctor about seeing a pain management specialist.

Ostomy care

If you have an ostomy, you may want to join an ostomy support group. Another option is to see a health care provider that specializes in ostomy care, such as an ostomy nurse.

People with ostomies can still live very active lifestyles. However, it's a good idea to talk to an ostomy professional before doing any intense physical activity.

Sexual health

Many people experience sexual side effects after chemotherapy and radiation. For those with a penis, this could mean difficulty getting an erection. Those with a vagina may experience vaginal dryness and pain during sex.

Radiation therapy to the pelvis can cause the vagina to become shorter and narrower. This is called vaginal stenosis. Vaginal stenosis can make it uncomfortable or even painful to have sex, or to have vaginal examinations by a doctor. Vaginal dilator therapy can help. A vaginal dilator is a device used to gradually stretch or widen the vagina.

The size of the dilator can be increased over time as the vagina lengthens and widens. Use of a vaginal dilator can begin as soon as you are healed after treatment.

Using a dilator during radiation treatment sessions can also help maintain sexual health after treatment.

Ostomy care

You can still live an active lifestyle with an ostomy. Joining a support group or seeing an ostomy nurse can help you learn more about living with an ostomy.



Early menopause

Menopausal hormone therapy

The ovaries make hormones needed for menstruation and natural pregnancy. Radiation damages the ovaries, causing hormone levels to drop. This is known as premature ovarian insufficiency (POI), or radiation-induced premature menopause. It causes infertility and may also cause symptoms of menopause, including:

- Stopping of periods
- Hot flashes
- Sleeping problems
- Night sweats
- Weight gain
- Changes in mood
- Vaginal dryness

Menopausal hormone therapy may help lessen these side effects. This approach used to be called hormone replacement therapy, or HRT.

One option is systemic combination therapy. Both estrogen and progestogen are given, usually either as a pill or a patch placed on the skin. The hormones travel throughout the body in the bloodstream. Progestogen is included because giving estrogen by itself can increase the risk of uterine cancer in those with a uterus.

Another option for menopausal hormone therapy is vaginal estrogen cream or tablets. This type may be the best option for symptoms that mainly affect the vagina, such as dryness.

Discussion with a specialized menopausal symptom team may be helpful to determine whether this treatment is right for you.

Paying for care

Cancer survivors face a unique financial burden. The cost of cancer care can become unmanageable, especially for those with little or no health insurance. You may also have costs not directly related to treatment, such as travel expenses and the cost of childcare.

The term financial toxicity is used to describe the problems patients face related to the cost of medical care. Financial toxicity can affect your quality of life and access to needed health care.

If you need help paying for cancer care, financial assistance may be available. Talk with a patient navigator, your care team's social worker, and your hospital's financial services department.

Healthy habits for cancer survivors



A low sugar (called glycemic load, or GL) diet may help prevent the return of rectal cancer. Low GL foods cause a slower and smaller rise in blood sugar levels than other carbohydrate-containing foods.



Talk to your doctor about the possible risks and benefits of long-term aspirin therapy to help prevent the return of colorectal cancers.



Follow your primary care provider's recommendations for screening for other cancers based on your age, sex, and risk level.



Get other recommended health care for your age, such as blood pressure checks, hepatitis C screening, and immunizations (like the flu shot).



Eat a diet rich in plant-based foods. Limit red meat and processed foods. Ask your provider for a referral to a cancer nutritionist.



Alcohol can increase the risk of some cancers. Drink little to no alcohol.



Move more, rest less. Try to exercise at a moderate intensity for at least 150 minutes per week.

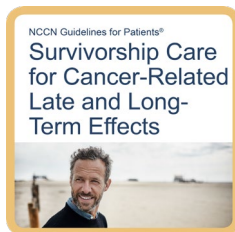
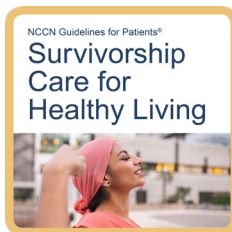


Maintain a healthy weight. Tracking your weight, food intake, and activity levels may help you meet your goals.

More information

For more information on cancer survivorship, the following are available at [NCCN.org/patientguidelines](https://www.nccn.org/patientguidelines) and on the [NCCN Patient Guides for Cancer](#) app:

- *Survivorship Care for Healthy Living*
- *Survivorship Care for Cancer-Related Late and Long-Term Effects*



Key points

- Survivorship describes the physical, emotional, and financial issues unique to cancer survivors. It starts at diagnosis and continues through your lifetime.
- A survivorship care plan is helpful in transitioning your care to your primary care doctor.
- If you have concerns about paying for your cancer care, financial help may be available.
- Keeping up with healthy habits can help prevent second cancers and other health problems.

Questions to ask

- Can you give me a survivorship care plan? How will you provide it?
- Does this treatment center have a survivorship care program?
- What can I expect during a survivorship care visit?



Take our survey and help make the NCCN Guidelines for Patients better for everyone!

[NCCN.org/patients/comments](https://www.nccn.org/patients/comments)

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Other resources

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Want to learn more? Here's how you can get additional help.

What else to know

This guide helps you know your options so you can make informed decisions and improve your cancer care. But it's not the only resource that you have.

Ask for as much information and help as you need. Many people are interested in learning more about:

- Managing side effects
- Genetic testing
- Building a support system
- Lifestyle changes after cancer
- Coping with other health problems

What else to do

Your health care center can help you with next steps. It often has on-site resources to help meet your needs and find answers to your questions. Health care centers can also inform you of resources in your community.

In addition to help from your providers, the resources listed in the next section provide support for many people like yourself.

Where to get help

Look through the list below and visit the provided websites to learn more about these organizations.

Anal Cancer Foundation
analcancerfoundation.org

Bone Marrow and Cancer Foundation
bonemarrow.org

CancerCare
cancercares.org

Cancer Hope Network
cancerhopenetwork.org

Cancer Survivor Care
cancersurvivorcare.org

Colorectal Cancer Alliance
ccalliance.org

Fight Colorectal Cancer
fightcolorectalcancer.org

GRACE
cancergrace.org

HPV Cancers Alliance
hpvalliance.org

Imerman Angels
imermanangels.org

Love Your Buns

loveyourbuns.org

My Faulty Gene

myfaultygene.org

Paltown Development Foundation

paltown.org

PAN Foundation

panfoundation.org

Triage Cancer

triagecancer.org

Questions to ask

- Who can I talk to about help with housing, food, and other basic needs?
- What help is available for transportation, childcare, and home care?
- Are there other services available to me and my caregivers?



We want your feedback!

Our goal is to provide helpful and easy-to-understand information on cancer.

Take our survey to let us know what we got right and what we could do better.

[NCCN.org/patients/feedback](https://nccn.org/patients/feedback)



Words to know

abdominoperineal resection (APR)

A type of transabdominal surgery used to remove tumors in the lower rectum. Involves removing the rectum and anus and creating a permanent colostomy for stool to leave the body.

ablation

A local therapy used to destroy tumors in the liver or lungs. Also called image-guided thermal ablation.

adenocarcinoma

Cancer that starts in the glands that line your organs. The most common type of rectal cancer.

adenoma

The most common type of colorectal polyp and the most likely to form cancer cells. Also called adenomatous polyp.

biomarkers

Specific features of cancer cells used to guide treatment. Biomarkers are often mutations (changes) in the DNA of the cancer cells.

CAPEOX

A chemotherapy regimen that includes capecitabine and oxaliplatin.

carcinoembryonic antigen (CEA)

A protein released by some tumors that can be detected in blood.

chemoembolization

Treatment for metastatic liver tumors that delivers spheres carrying chemotherapy to the liver vessels supplying the tumor.

colon

The first and longest section of the large bowel. Unused food is turned into stool in the colon.

colostomy

Surgery to connect a part of the colon to the outside of the abdomen and that allows stool to drain into a bag.

endoscopic submucosal dissection (ESD)

A minimally invasive type of surgery used to remove small rectal tumors in one piece. Also called endoscopic mucosal resection (EMR).

external beam radiation therapy (EBRT)

Treatment with high-energy rays received from a machine outside the body.

familial adenomatous polyposis (FAP)

An inherited cancer syndrome that increases the risk of colorectal cancer.

FOLFIRI

A chemotherapy regimen used for some advanced colorectal cancers. Includes leucovorin calcium, fluorouracil, and irinotecan.

FOLFIRINOX

A chemotherapy regimen used for some colorectal cancers. Includes leucovorin calcium (folinic acid), fluorouracil, irinotecan, and oxaliplatin.

FOLFOX

A chemotherapy regimen that includes leucovorin calcium, fluorouracil, and oxaliplatin.

interventional oncology/radiology

A medical specialty that uses imaging techniques to deliver minimally invasive cancer treatments.

large intestine (bowel)

A long, tube-shaped organ that forms the last part of the digestive system. Includes the colon, rectum, and anus.

lymph

A clear fluid containing white blood cells.

low anterior resection (LAR)

A type of transabdominal surgery for removing tumors in the middle or upper rectum. Involves removing the rectal tumor and part or all of the last section of the colon. The colon is connected to the remaining rectum (when possible) or attached directly to the anus.

lymph node

Small groups of disease-fighting cells located throughout the body.

metastasis

The spread of cancer cells from the first (primary) tumor to a distant site.

mismatch repair deficiency (dMMR)/high microsatellite instability (MSI-H)

A biomarker (feature) of some colorectal cancers that is used to guide treatment. All colorectal cancers should be tested for this biomarker.

mismatch repair proficient (pMMR)/microsatellite stable (MSS)

Describes cancers that don't have the mismatch repair deficiency (dMMR) biomarker or the microsatellite instability-high (MSI-H) biomarker.

pathologist

A doctor who specializes in testing cells and tissue to find disease.

polyp

An overgrowth of cells on the inner lining of the colon wall. Pedunculated polyps stick out from the rectal wall, while sessile polyps are flatter.

portal vein embolization

A minimally invasive procedure to enlarge the liver. May be needed if a liver resection is planned.

radioembolization

A treatment for liver tumors that uses radioactive spheres injected in the hepatic (liver) artery (the blood vessel supplying tumor). Also called selective internal radiation therapy, or SIRT.

rectum

An organ in the digestive system that holds stool until it exits the body through the anus.

recurrence

The return of cancer after a cancer-free period.

resectable

Describes a cancer that can be safely removed using surgery.

stereotactic body radiation therapy (SBRT)

A highly specialized type of radiation therapy. May be used to treat rectal cancer that has spread to the liver, lungs, or bone.

supportive care

A range of resources and care to improve quality of life for cancer survivors.

systemic therapy

The use of medicines that enter the bloodstream. Chemotherapy, targeted therapy, and immunotherapy are types of systemic therapy.

unresectable

Describes a cancer that can't be safely removed using surgery.

NCCN Contributors

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NCCN Cancer Centers

For contact information, visit [NCCN.org/cancercenters](https://www.nccn.org/cancercenters).

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Cleveland, Ohio

City of Hope National Medical Center
Duarte, California

Dana-Farber Cancer Institute
Boston, Massachusetts

Duke Cancer Institute
Durham, North Carolina

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Philadelphia, Pennsylvania

Fred & Pamela Buffett Cancer Center
Omaha, Nebraska

Fred Hutchinson Cancer Center
Seattle, Washington

Huntsman Cancer Institute at the University of Utah
Salt Lake City, Utah

Indiana University Melvin and Bren Simon
Comprehensive Cancer Center
Indianapolis, Indiana

Johns Hopkins Kimmel Cancer Center
Baltimore, Maryland

Mass General Brigham Cancer Institute
Boston, Massachusetts

Mayo Clinic Comprehensive Cancer Center
Phoenix/Scottsdale, Arizona
Jacksonville, Florida
Rochester, Minnesota

Memorial Sloan Kettering Cancer Center
New York, New York

Moffitt Cancer Center
Tampa, Florida

O'Neal Comprehensive Cancer Center at UAB
Birmingham, Alabama

Robert H. Lurie Comprehensive Cancer Center
of Northwestern University
Chicago, Illinois

Roswell Park Comprehensive Cancer Center
Buffalo, New York

Siteman Cancer Center at Barnes-Jewish Hospital
and WashU Medicine
St. Louis, Missouri

St. Jude Children's Research Hospital/
The University of Tennessee Health Science Center
Memphis, Tennessee

Stanford Cancer Institute
Stanford, California

The Ohio State University Comprehensive Cancer Center -
James Cancer Hospital and Solove Research Institute
Columbus, Ohio

The UChicago Medicine Comprehensive Cancer Center
Chicago, Illinois

The University of Texas MD Anderson Cancer Center
Houston, Texas

UC Davis Comprehensive Cancer Center
Sacramento, California

UC San Diego Moores Cancer Center
La Jolla, California

UCLA Jonsson Comprehensive Cancer Center
Los Angeles, California

UCSF Helen Diller Family Comprehensive Cancer Center
San Francisco, California

University of Colorado Cancer Center
Aurora, Colorado

University of Michigan Rogel Cancer Center
Ann Arbor, Michigan

University of Wisconsin Carbone Cancer Center
Madison, Wisconsin

UT Southwestern Simmons
Comprehensive Cancer Center
Dallas, Texas

Vanderbilt-Ingram Cancer Center
Nashville, Tennessee

Yale Cancer Center/Smilow Cancer Hospital
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